

2.3 Properties of Determinants; Cramer's Rule

In this section we develop fundamental properties of determinants and use them to derive a formula for the inverse of an invertible matrix and formulas for solving certain linear systems.

Basic Properties of Determinants

For $n \times n$ matrices A and B and any scalar k , we consider possible relationships among $\det(kA)$, $\det(A + B)$, and $\det(AB)$.

Since a common factor from any row can be moved through the determinant sign, and kA has k as a common factor in each of its n rows:

$$\boxed{\det(kA) = k^n \det(A)} \quad (1)$$

For example:

$$\begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ ka_{21} & ka_{22} & ka_{23} \\ ka_{31} & ka_{32} & ka_{33} \end{vmatrix} = k^3 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

No simple relationship exists among $\det(A)$, $\det(B)$, and $\det(A + B)$; in particular, $\det(A + B)$ usually does *not* equal $\det(A) + \det(B)$.

EXAMPLE 1 | $\det(A + B) \neq \det(A) + \det(B)$

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, so $A + B = \begin{bmatrix} 4 & 3 \\ 3 & 8 \end{bmatrix}$. Then $\det(A) = 1$, $\det(B) = 8$, $\det(A + B) = 23$; thus $\det(A + B) \neq \det(A) + \det(B)$.

There *is* a useful relationship when matrices differ in exactly one row or column:

Theorem 2.3.1

Let A , B , and C be $n \times n$ matrices that differ only in a single row (say the r th), with the r th row of C equal to the sum of the corresponding entries of A and B . Then

$$\det(C) = \det(A) + \det(B)$$

The same result holds for columns.

EXAMPLE 2 | Sums of Determinants

Verify the following equality by evaluating each determinant:

$$\det \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 1+0 & 4+1 & 7+(-1) \end{bmatrix} = \det \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 1 & 4 & 7 \end{bmatrix} + \det \begin{bmatrix} 1 & 7 & 5 \\ 2 & 0 & 3 \\ 0 & 1 & -1 \end{bmatrix}$$

Determinant of a Matrix Product

Despite the complexity of matrix multiplication, a surprisingly simple relationship holds:

$$\det(AB) = \det(A) \det(B) \quad (2)$$

We prove this by first establishing the following lemma.

Lemma 2.3.2

If B is an $n \times n$ matrix and E is an $n \times n$ elementary matrix, then

$$\det(EB) = \det(E) \det(B)$$

Proof. Three cases corresponding to the three ERO types:

Case 1. E results from multiplying row i of I_n by k . Then EB has row i of B scaled by k , so $\det(EB) = k \det(B)$ (Theorem 2.2.3(a)). Also $\det(E) = k$ (Theorem 2.2.4(a)), so $\det(EB) = \det(E) \det(B)$.

Cases 2 and 3. The proofs for the row-interchange and row-addition cases follow the same pattern and are left as exercises. ■

Remark. By repeated application of Lemma 2.3.2, if $E_1, E_2, \dots, E_r$ are $n \times n$ elementary matrices and B is $n \times n$:

$$\det(E_1 E_2 \cdots E_r B) = \det(E_1) \det(E_2) \cdots \det(E_r) \det(B) \quad (3)$$

Determinant Test for Invertibility

Theorem 2.3.3

A square matrix A is invertible if and only if $\det(A) \neq 0$.

Proof. Let R be the reduced row echelon form of A , and let $E_1, \dots, E_r$ be the elementary matrices such that $R = E_r \cdots E_2 E_1 A$. From (3):

$$\det(R) = \det(E_r) \cdots \det(E_2) \det(E_1) \det(A) \quad (4)$$

Since $\det(E_i) \neq 0$ for each i , $\det(A)$ and $\det(R)$ are either both zero or both nonzero.

- If A is invertible $\Rightarrow R = I$ (Thm. 1.6.4) $\Rightarrow \det(R) = 1 \neq 0 \Rightarrow \det(A) \neq 0$.
- If $\det(A) \neq 0 \Rightarrow \det(R) \neq 0 \Rightarrow R$ has no zero row (Thm. 1.4.3) $\Rightarrow R = I \Rightarrow A$ invertible (Thm. 1.6.4). ■

EXAMPLE 3 | Determinant Test for Invertibility

Since the first and third rows of $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 4 & 6 \end{bmatrix}$ are proportional, $\det(A) = 0$. Thus A is not invertible.

Theorem 2.3.4

If A and B are square matrices of the same size, then

$$\det(AB) = \det(A) \det(B)$$

Proof. **Case 1.** A not invertible $\Rightarrow AB$ not invertible (Thm. 1.6.5) $\Rightarrow \det(AB) = 0$ and $\det(A) = 0$, so $\det(AB) = \det(A) \det(B) = 0$.

Case 2. A invertible $\Rightarrow A = E_1 E_2 \cdots E_r$ (Thm. 1.6.4), so $AB = E_1 \cdots E_r B$. Applying (3):

$$\det(AB) = \det(E_1) \cdots \det(E_r) \det(B) = \det(E_1 E_2 \cdots E_r) \det(B) = \det(A) \det(B). \blacksquare$$

EXAMPLE 4 | Verifying $\det(AB) = \det(A) \det(B)$

Let $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 \\ 5 & 8 \end{bmatrix}$, $AB = \begin{bmatrix} 2 & 17 \\ 3 & 14 \end{bmatrix}$. Then

$$\det(A) = 1, \quad \det(B) = -23, \quad \det(AB) = -23 = \det(A) \det(B). \quad \checkmark$$

Theorem 2.3.5

If A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof. Since $A^{-1}A = I$, we have $\det(A^{-1}) \det(A) = \det(I) = 1$, so $\det(A^{-1}) = 1/\det(A)$. $\blacksquare$

Adjoint of a Matrix

In the cofactor expansion of $\det(A)$ along row i , multiplying entries of row i by the cofactors of a *different* row $j \neq i$ always yields zero (and likewise for columns).

EXAMPLE 5 | Entries and Cofactors from Different Rows

Let $A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$. The cofactors of A are:

$$C_{11} = 12, \quad C_{12} = 6, \quad C_{13} = -16$$

$$C_{21} = 4, \quad C_{22} = 2, \quad C_{23} = 16$$

$$C_{31} = 12, \quad C_{32} = -10, \quad C_{33} = 16$$

Cofactor expansion along row 1: $\det(A) = 3(12) + 2(6) + (-1)(-16) = 36 + 12 + 16 = 64$.

Cofactor expansion along column 1: $\det(A) = 3(12) + 1(4) + 2(12) = 64$. $\checkmark$

Row 1 entries $\times$ row 2 cofactors: $3C_{21} + 2C_{22} + (-1)C_{23} = 12 + 4 - 16 = 0$.

Column 1 entries $\times$ column 2 cofactors: $3C_{12} + 1C_{22} + 2C_{32} = 18 + 2 - 20 = 0$.

Definition 1

If A is any $n \times n$ matrix and C_{ij} is the cofactor of a_{ij} , then the **matrix of cofactors** from A is

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

The transpose of this matrix is called the **adjoint** of A , denoted $\text{adj}(A)$.

EXAMPLE 6 | Adjoint of a 3×3 Matrix

For $A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$ (from Example 5), the matrix of cofactors is $\begin{bmatrix} 12 & 6 & -16 \\ 4 & 2 & 16 \\ 12 & -10 & 16 \end{bmatrix}$, so

$$\text{adj}(A) = \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{bmatrix}$$

Theorem 2.3.6 — Inverse of a Matrix Using Its Adjoint

If A is an invertible $n \times n$ matrix, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) \quad (6)$$

Proof. The (i, j) entry of $A \text{adj}(A)$ is:

$$a_{i1}C_{j1} + a_{i2}C_{j2} + \cdots + a_{in}C_{jn} \quad (7)$$

If $i = j$: this is the cofactor expansion of $\det(A)$ along row i . If $i \neq j$: entries from row i multiply cofactors from row j , giving 0 (Example 5). Therefore:

$$A \text{adj}(A) = \begin{bmatrix} \det(A) & 0 & \cdots & 0 \\ 0 & \det(A) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \det(A) \end{bmatrix} = \det(A) \cdot I \quad (8)$$

Since $\det(A) \neq 0$, we have $A \left[\frac{1}{\det(A)} \text{adj}(A) \right] = I$, so $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$. ■

EXAMPLE 7 | Using the Adjoint to Find an Inverse Matrix

Use Formula (6) to find the inverse of A from Example 6.

SOLUTION: From Example 5, $\det(A) = 64$. Thus:

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) = \frac{1}{64} \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{bmatrix} = \begin{bmatrix} \frac{12}{64} & \frac{4}{64} & \frac{12}{64} \\ \frac{6}{64} & \frac{2}{64} & -\frac{10}{64} \\ -\frac{16}{64} & \frac{16}{64} & \frac{16}{64} \end{bmatrix}$$

Cramer's Rule

Theorem 2.3.7 — Cramer's Rule

If $A\mathbf{x} = \mathbf{b}$ is a system of n equations in n unknowns with $\det(A) \neq 0$, the unique solution is

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \quad \dots, \quad x_n = \frac{\det(A_n)}{\det(A)}$$

where A_j is the matrix obtained by replacing the j th column of A by $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$.

Proof. Since $\det(A) \neq 0$, A is invertible and $\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det(A)} \operatorname{adj}(A)\mathbf{b}$. The j th entry is:

$$x_j = \frac{b_1 C_{1j} + b_2 C_{2j} + \dots + b_n C_{nj}}{\det(A)} \quad (9)$$

Since A_j differs from A only in column j , the cofactors of that column in A_j equal $C_{1j}, \dots, C_{nj}$ of A . Hence $\det(A_j) = b_1 C_{1j} + \dots + b_n C_{nj}$, giving $x_j = \det(A_j) / \det(A)$. ■

EXAMPLE 8 | Using Cramer's Rule to Solve a Linear System

Use Cramer's rule to solve:

$$x_1 + 2x_3 = 6, \quad -3x_1 + 4x_2 + 6x_3 = 30, \quad -x_1 - 2x_2 + 3x_3 = 8$$

SOLUTION:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -3 & 4 & 6 \\ -1 & -2 & 3 \end{bmatrix}, \quad A_1 = \begin{bmatrix} 6 & 0 & 2 \\ 30 & 4 & 6 \\ 8 & -2 & 3 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 6 & 2 \\ -3 & 30 & 6 \\ -1 & 8 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 1 & 0 & 6 \\ -3 & 4 & 30 \\ -1 & -2 & 8 \end{bmatrix}$$

Therefore:

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{-40}{44} = -\frac{10}{11}, \quad x_2 = \frac{\det(A_2)}{\det(A)} = \frac{72}{44} = \frac{18}{11}, \quad x_3 = \frac{\det(A_3)}{\det(A)} = \frac{152}{44} = \frac{38}{11}$$

Equivalence Theorem

By merging Theorem 2.3.3 with the earlier Theorem 1.6.4, we obtain:

Theorem 2.3.8 — Equivalent Statements

If A is an $n \times n$ matrix, the following statements are equivalent:

- (a) A is invertible.
- (b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- (c) The reduced row echelon form of A is I_n .
- (d) A can be expressed as a product of elementary matrices.
- (e) $A\mathbf{x} = \mathbf{b}$ is consistent for every $n \times 1$ matrix $\mathbf{b}$.
- (f) $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $n \times 1$ matrix $\mathbf{b}$.
- (g) $\det(A) \neq 0$.

Exercise Set 2.3

In Exercises 1–3, verify that $\det(kA) = k^n \det(A)$.

1. $A = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}; k = 2$

3. $A = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix}; k = -2$

In Exercise 5, verify that $\det(AB) = \det(BA)$ and determine whether $\det(A+B) = \det(A) + \det(B)$.

5. $A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 & 3 \\ 7 & 1 & 2 \\ 5 & 0 & 1 \end{bmatrix}$

In Exercises 7–14, use determinants to decide whether the matrix is invertible.

7. $A = \begin{bmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{bmatrix}$

9. $A = \begin{bmatrix} 2 & -3 & 5 \\ 0 & 1 & -3 \\ 0 & 0 & 2 \end{bmatrix}$

11. $A = \begin{bmatrix} 4 & 2 & 8 \\ -2 & 1 & -4 \\ 3 & 1 & 6 \end{bmatrix}$

14. $A = \begin{bmatrix} \sqrt{2} & -\sqrt{7} & 0 \\ 3\sqrt{2} & -3\sqrt{7} & 0 \\ 5 & -9 & 0 \end{bmatrix}$

In Exercises 15–17, find the values of k for which the matrix A is invertible.

15. $A = \begin{bmatrix} k-3 & -2 \\ -2 & k-2 \end{bmatrix}$

17. $A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 6 \\ k & 3 & 2 \end{bmatrix}$

In Exercises 19 and 22, decide whether the matrix is invertible; if so, use the adjoint method to find its inverse.

19. $A = \begin{bmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{bmatrix}$

22. $A = \begin{bmatrix} 2 & 0 & 0 \\ 8 & 1 & 0 \\ -5 & 3 & 6 \end{bmatrix}$

In Exercises 24–27, solve by Cramer's rule, where it applies.

24. $7x_1 - 2x_2 = 3, \quad 3x_1 + x_2 = 5$

25. $4x + 5y = 2, \quad 11x + y + 2z = 3, \quad x + 5y + 2z = 1$

27. $x_1 - 3x_2 + x_3 = 4, \quad 2x_1 - x_2 = -2, \quad 4x_1 - 3x_3 = 0$

30. Show that $A = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is invertible for all values of θ ; then find A^{-1} using Theorem 2.3.6.

33. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ with $\det(A) = -7$. Find:

(a) $\det(3A)$

(b) $\det(A^{-1})$

(c) $\det(2A^{-1})$

(d) $\det((2A)^{-1})$

(e) $\det \begin{bmatrix} a & g & d \\ b & h & e \\ c & i & f \end{bmatrix}$

34. In each part, find the determinant given that A is a 4×4 matrix with $\det(A) = -2$:

(a) $\det(-A)$

(b) $\det(A^{-1})$

(c) $\det(2A^T)$

(d) $\det(A^3)$